

Reliability, Validity, and Ethics in Sports Analytics

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HES 3000-002, Sport Performance Analytics

Illusion of precision

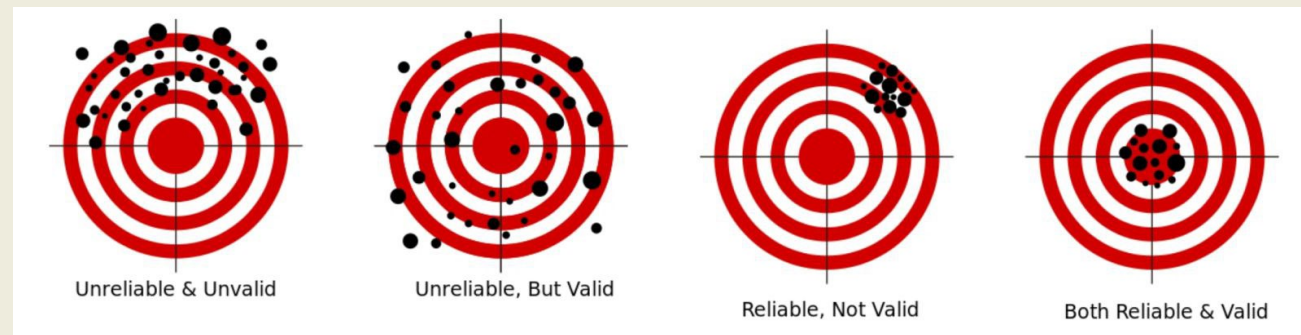
- Media framing implies that GPS vests, force plates, and optical tracking systems give us an infallible, objective window into an athlete's soul and body.
- Systems may produce numbers with multiple decimal places, but precision does not guarantee accuracy.

Messy data

- Sport data are affected by:
 - Inconsistent procedures
 - Human error
 - Environmental conditions
 - Missing observations
- Before analyzing performance, we must understand how the data were produced.

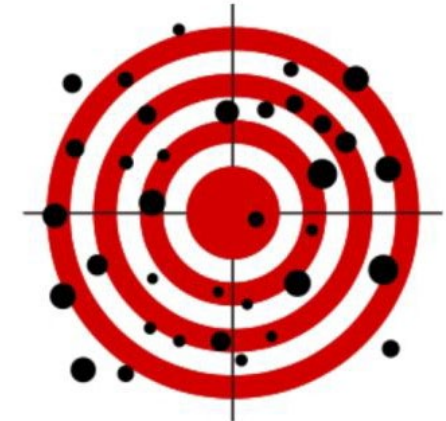
Trust in the measurement

Concept	Primary question
Reliability	Can the measurement be reproduced consistently?
Validity	Does the measurement represent what we intend to measure?



Reliability = Consistency

- A reliable measurement produces similar results when the underlying performance has not meaningfully changed
- Potential sources of inconsistency include:
 - Device measurement error
 - Device-to-device differences
 - Rater or analyst differences
 - Inconsistent testing procedures
 - Normal biological variation



Unreliable, But Valid



Both Reliable & Valid

Types of reliability

- Test–retest reliability
 - Does the same athlete produce similar results when the test is repeated under similar conditions?
 - Eg: 1RM
- Intra-rater reliability
 - Does the same observer produce similar measurements when repeating the process?
 - Eg: same statistician quantifying player stats
- Inter-rater reliability
 - Do different observers produce similar measurements?
 - Eg: multiple statisticians quantifying player stats

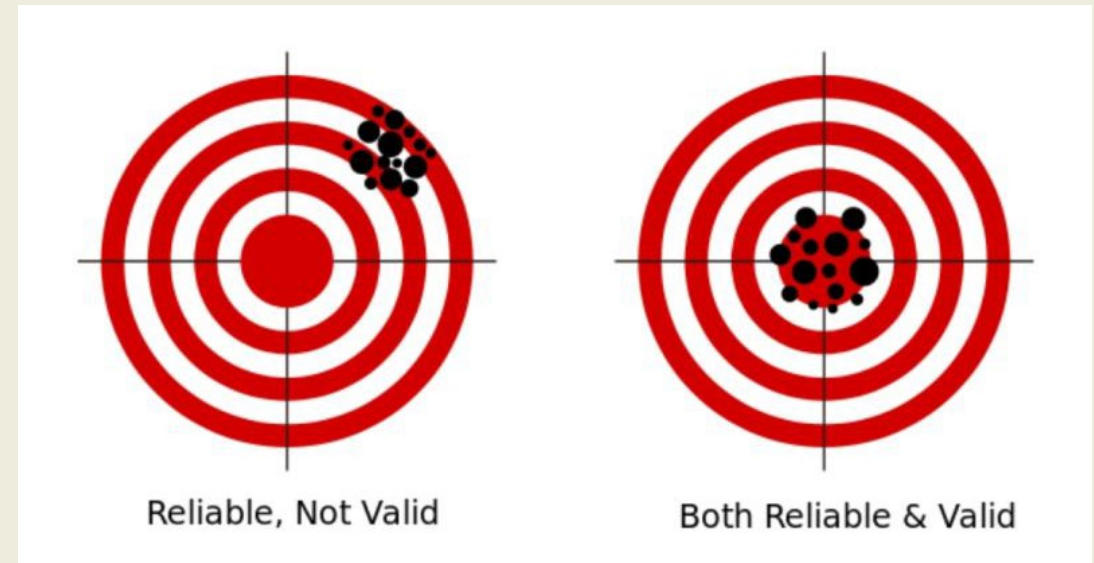
How can we make sure data is reliable?

Reliable... but valid?

- An athlete completes the same jump test three times under similar conditions:
 - Trial 1: 19.2 in
 - Trial 2: 19.5 in
 - Trial 3: 19.4 in
- Is our data reliable?
 - appears highly reliable.
- This does not tell us whether the device is measuring jump height correctly.

Validity = Measuring What We Intend

- A valid measurement accurately represents the concept or outcome we want to study.
- Examples:
 - Does the device accurately measure distance traveled?
 - Does “player load” represent overall physical stress?
 - Does AVG represent offensive contribution?



Types of Validity

- Criterion validity
 - Does the measurement agree with an accepted reference method?
 - Eg: jump height
- Construct validity
 - Does the measurement behave as expected based on the underlying concept?
 - Eg: RPE vs actual intensity
- Ecological validity
 - Does the measurement represent performance in the real sporting environment?
 - Eg: lab test results vs on-field performance

Reliable but Not Valid

Imagine a bathroom scale that always reports weight 5 kg too high.

- It gives consistent measurements → reliable
 - It does not give accurate measurements → not valid.
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- In sport, a metric may consistently measure **something** without measuring what coaches think it measures.
 - Is this a problem?

Understanding What the Metric Actually Means

Consider “player load.”

- A device may calculate player load consistently, but:
 - Does it measure total physiological stress?
 - Does it account for collisions?
 - Does it capture psychological fatigue?
 - Can values be compared across devices or manufacturers?
 - Does the same value mean the same thing for every athlete?

Measurement Error Is Not Athlete Change

- Observed values contain multiple components:

Observed value = true performance + measurement error + normal variation

- A change in the number does not automatically indicate a **meaningful change** in the athlete.
 - Is the change larger than typical measurement error?
 - Is it larger than the athlete's normal day-to-day variation?
 - Was the measurement collected under comparable conditions?

We will revisit this when discussing analytics in athlete monitoring

Garbage In, Garbage Out

- Your analysis is only as good as your data source
- Stats and models cannot fix flawed data



Sensor Audit

- Think of a wearable or performance-monitoring device you have used. **What is one factor that could produce misleading or unusable data?**
- Possible categories:
 - Device placement
 - Poor contact
 - Environment
 - Athlete behavior

Sensor Audit Example

- Poor contact or improper placement of a heart-rate chest strap can produce:
 - Missing observations
 - Sudden heart-rate spikes
 - Delayed responses
 - Data that appear real but are physiologically unlikely



Protect your analysis from bad data

- Before analyzing a dataset, ask:
 - Who collected the data, and why?
 - What technology or method was used?
 - Has that method been validated for this setting?
 - Was the methodology consistent across athletes and over time?
 - What values are missing?
 - Who or what is not represented?
 - What decisions could be affected by the analysis?
 - What can the data not establish?

Data-Quality Red Flags

- Watch for:
 - Impossible or implausible values
 - Sudden changes in measurement scale
 - Duplicate observations
 - Unexplained missing data
 - Changes in devices or software
 - Inconsistent testing conditions
 - Different definitions across seasons
 - Metrics with unclear calculations
 - Systematic differences between groups
 - Values that are unusually perfect

Ethics in Sports Analytics

Why does it matter?

- Athlete data may influence:
 - Playing time
 - Team selection
 - Scholarships
 - Contracts
 - Injury management
 - Return-to-play decisions
 - How coaches perceive an athlete

consent and ownership

- Athlete monitoring raises several questions:
 - Do athletes understand what is being collected?
 - Can they meaningfully refuse?
 - Who owns the data?
 - Who can access it?
 - How long will it be stored?
 - Can it be used for purposes beyond the original reason for collection?
 - What happens when an athlete leaves the team?
- **Participation is not necessarily the same as informed consent.**

weaponization of data

- Data collected to support athletes can also be used against them.
 - Using wellness responses to question commitment
 - Using monitoring data during contract negotiations
 - Punishing athletes for reporting fatigue or pain
 - Sharing data outside its original purpose
- If athletes believe honesty will be punished, they may stop providing honest data.
 - This produces both an ethical failure and a data-quality failure.

Establishing Trust

- Educate athletes and coaches about what is collected, what the metrics mean, and how the data may be used.
- Limit data access to people who need it.
- Define acceptable uses before collecting the data.
- Establish rules for storage, sharing, and deletion.

The resource divide

- Funding determines ability to collect data
 - Tracking technology
 - Staff and technical expertise
 - Time for quality control
- Better-resourced organizations do not simply have more data. They often have more opportunities to make the data usable.
- How does this affect:
 - The quality of analytics?
 - Outcomes in each domain?

How Does the Resource Divide Affect Analytics?

- Lower-resource programs may face:
 - Smaller datasets
 - Less complete dataset
 - Less staff time for cleaning and interpretation

Domain	Possible consequence
Athlete monitoring	Less ability to individualize training or detect meaningful changes
Scouting	Greater reliance on visible athletes and traditional networks
Strategy	Less detailed opponent and game analysis

The invisible athlete

- An athlete may be overlooked because they are:
 - Outside major recruiting networks
 - Playing in a lower-resource environment
 - Missing from commercial databases
 - Competing against weaker or poorly documented opponents
 - Measured using inconsistent methods
 - Strong in qualities that the available metrics do not capture
- No data does not mean no ability.
 - It may only mean that the athlete was never observed by the system.

Making the invisible athlete

Statistical models can only identify athletes represented by their data.

- If scouting data primarily include:
 - Highly recruited athletes
 - Major competitions
 - Well-funded programs
 - Easily measured performance outcomes
- then the resulting analysis may reinforce existing scouting patterns.

A model may learn who traditionally receives opportunities, not who has the most potential.

Finding the invisible athlete

- Suppose the original question is:
 - Which athletes have the potential to succeed at the next level?
- The available data might only allow us to answer:
 - Which highly visible athletes currently have the best statistics?
- Those are not the same question.
 - Changing the question would be easier but it would preserve the same blind spots as everyone else.

Streetlight effect - revisited

- The poorly illuminated area may contain the greatest competitive advantage.
- Response to limited data:

Streetlight response	Appropriate response
Replace it with what the dataset can easily answer	Preserve the purpose of the question
Analyze only familiar variables	Seek new data or useful proxies
Overstate what available data establish	State uncertainty and limitations
Reproduce existing rankings	Investigate overlooked athletes

Pursuing the harder question

- Overlooked athletes may be:
 - Less expensive to recruit or acquire
 - Undervalued by competitors
 - Available later in a draft or recruiting process
 - Better than their conventional statistics suggest
- If every organization uses the same convenient data, they will tend to identify the same athletes.

Alternative data and methods

- When the available data do not answer the original question, analysts can:
 - Seek alternative data sources
 - Collect new measurements
 - Develop proxies for qualities that are difficult to observe
- The goal is not to force an answer from inadequate data. It is to find a better way to investigate the original question.

Summary & Looking Ahead

- Data are not automatically objective simply because they are numerical. Their credibility depends on how they were collected, interpreted, and used.
- Before analysis, consider **reliability**, **validity** and **ethics**.
- Next: data characteristics